Semidefinite Programming and Scheduling

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Abstract. We explore semidefinite programming relaxations for a variety of mutual-exclusion and timetabling problems due to Marecek and Parkes [36]. We examine an algorithm is which solves these relaxations and acknowledge some methods to recover an actual assignment. Finally, we look at practical performance metrics on standard timetabling input.

Keywords: Semidefinite Programming · Bounded Vertex-Colouring · Mutual-Exclusion Scheduling.

1 Introduction

Since Goemans and Williams' use of semidefinite programming (SDP) to obtain a .87856-approximation of the APX-hard MaxCut problem [22], SDP has rapidly gained popularity for use in combinatorial optimization. For instance, Wolkowicz et al. and Ajos et al. showed how to use SDP to obtain lower bounds [47, 3], while Lau et al. used randomized rounding to obtain good solutions [30]. In their paper, Marecek and Parkes explore the application of SDP towards practical scheduling and timetabling problems.

Mutual-exclusion constraints in the context of scheduling and timetabling problems stipulate that certain pairs of tasks or events cannot be executed simultaneously. This corresponds to the VertexColouring problem in the *conflict graph*, where every task or event is a node and two nodes are adjacent if they cannot happen at the same time. Marecek and Parkes explore the limits of SDP in practical settings where additional constraints are necessary. The authors show that SDP relaxations for problems of the described type can be formulated, starting with m-BoundedColouring of the conflict graph.

This report is organized as follows: Following some preliminaries, we present the SDP relaxations due to Marecek and Parkes in Section 2. In Section 3, we illustrate the key ideas of an algorithm which solves the relaxations as well as references for two algorithms for rounding the fractional solutions. Finally, in Section 4, we present results from the authors' computational experiments on conflict graphs from the International Timetabling Competition 2007 [7], the University of Toronto benchmark [16], as well as on random graphs.

1.1 Semidefinite Programming

Recall that a *semidefinite program* is a convex optimization problem over the semidefinite cone [8, 47, 3]. We write $\mathbb{S}^n$ to denote the vector space of symmetric

$n \times n$ matrices and $\mathbb{S}^n_+$ to denote the semidefinite cone. For $A, B \in \mathbb{S}^n$, $A \succeq B$ denotes $A - B \in \mathbb{S}^n_+$. Following common notation, we write $\lambda_1(A), \lambda_2(A), \ldots, \lambda_n(A)$ to denote the eigenvalues of A in non-increasing order. It is a well-known fact that $A \succeq B$ implies that $\lambda_i(A) \geq \lambda_i(B)$ for all $i \in [n]$.

Given input data $C \in \mathbb{S}^n, b \in \mathbb{R}^m$ and a linear map $\mathscr{A} : \mathbb{S}^n \to \mathbb{R}^m$, a popular standard form of a semidefinite program can expressed as:

$$\begin{aligned} \min \langle C, X \rangle & \qquad (P) \\ \mathscr{A}(X) &= b \\ X &\succeq 0 \end{aligned}$$

with its dual being

$$\begin{aligned} \max b^T y & \qquad (D) \\ \mathscr{A}^*(y) + S &= C \\ S &\succeq 0 \end{aligned}$$

Semidefinite programming can be seen as a strict generalization of linear programming. However, unlike linear programming, semidefinite programs can only be solved approximately, for instance, with an interior-point method [2].

1.2 Related Work

Recall the VertexColouring problem in an undirected graph $G = (V, E)$. The *chromatic number* of G, denoted $\chi(G)$, is the smallest possible number of colour classes. In the context of conflict graphs, each colour class is a set of tasks or events which can happen at the same time.

We say an algorithm is *efficient* if its running time is bounded above by a polynomial with respect to the input size. Using convex relaxations of the discrete colouring problem, one can efficiently compute lower bounds on $\chi(G)$ [46]. For one such lower bound, recall the *Lovasz theta number* $\theta(G)$. It is well-known that $\theta(G)$ is at least the size of the largest clique but at most the chromatic number [34]. Moreover, $\theta(G)$ can be efficiently computed to any accuracy by solving the following semidefinite program.

$$\begin{aligned} \theta(G) &= \max \langle J, X \rangle \\ \langle I, X \rangle &= 1 \\ X_{ij} &= 0 & \forall ij \in \bar{E} \\ X &\succeq 0 \end{aligned}$$

where J is the all-ones matrix of appropriate dimension. Other known lower bounds include the *strict vector chromatic number* [28]:

$$\begin{aligned} \theta_{\frac{1}{2}}(G) &= \max\langle J, X\rangle \\ \langle I, X\rangle &= 1 \\ X_{ij} &= 0 && \forall ij \in \bar{E} \\ X_{ij} &\geq 0 && \forall ij \in E \\ X &\succeq 0 \end{aligned}$$

as well as the *strong vector chromatic number* [43]:

$$\begin{aligned} \theta_2(G) &= \max\langle J, X\rangle \\ \langle I, X\rangle &= 1 \\ X_{ij} &\leq 0 && \forall ij \in \bar{E} \\ X &\succeq 0 \end{aligned}$$

Let $\omega(G)$ denote the *clique number* of G, that is, the size of the largest clique in G. Szegedy showed that these bounds form a hierarchy [43]:

$$\omega(G) \leq \theta_{\frac{1}{2}}(G) \leq \theta(G) \leq \theta_2(G) \leq \chi(G).$$

It is worth it to note that to the best knowledge of Marecek and Parkes, none of the formulations above has been extended to the case of m-BoundedColouring, where the size of each colour is at most some given constant.

Interestingly, the SDP relaxations of the MaxCut problem have been adapted to scheduling workload on two machines [41, 48] as well as home-away patterns in sports scheduling [42]. This idea has been further extended to scheduling with weighted completion time objectives on any number of unrelated machines. The basis for this is a clever rounding technique involving lift-and-project relaxations [5]. Bansal et al. also showed that the relaxation of Skutella [41] has a $\frac{3}{2}$ integrality gap.

2 SDP Formulations

In their paper, Marecek and Parkes developed a versatile framework of SDP relaxations applied towards scheduling. A straightforwards interpretation of a matrix variable $X \in \mathbb{S}^n$ is to think of

$$X_{ij} = \begin{cases} 1, & i, j \text{ are in the same colour class} \\ 0, & \text{else} \end{cases}$$

Let us refer to such a 0-1 symmetric matrix as an *integral scheduling matrix*.

Proposition 1 ([36]). *For any integral scheduling matrix, there is a permutation matrix P such that*

$$PXP^T = \begin{bmatrix} J_{d_1} & 0 & \dots & 0 \\ 0 & J_{d_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & J_{d_s} \end{bmatrix} = \oplus_{i=1}^{s} J_{d_i}$$

where J_d is the $d \times d$ matrix of all ones and $\sum_{k=1}^{s} d_k = n$.

As it turns out, we can completely characterize matrices which satisfy Proposition 1. Recall $\operatorname{diag} : \mathbb{S}^n \to \mathbb{R}^n$ is the linear map which extracts the diagonal of a matrix and $\bar{e}$ denotes the all-ones vector of appropriate dimension.

Proposition 2 ([19]). *For any 0-1 matrix $X \in \mathbb{S}^n$, there is a permutation matrix P such that PXP^T is a direct sum of s all-ones matrices if and only if $\operatorname{diag}(X) = \bar{e}$, $\operatorname{rank} X = s$ and M is positive semidefinite.*

In order to compute the chromatic colour, The results above suggest a non-convex relaxation based on minimizing the rank among all integral symmetric matrices satisfying the conditions in Proposition 2. All of Marecek and Parkes' SDP relaxations are based on this idea.

2.1 Mutual-Exclusion Scheduling

In one of the prototypical problems in scheduling, we need to assign n unit-time tasks, some of which cannot happen simultaneously, such that the number of time units required is minimal. Due to scarcity of resources, we also wish to restrict the total number of ongoing tasks at any given time to at most a constant m [44, 14, 4]. This problem can be elegantly formulated as the m-BoundedColouring problem within the conflict graph.

Problem 1 (m-Bounded Colouring). Given an undirected graph $G = (V, E)$ and $m \in \mathbb{Z}_+$, return a proper vertex-colouring such that each colour class has size at most m.

This problem has been previously explore as scheduling of unit-length tasks on m parallel machines with renewable resources [9], mutual-exclusion scheduling [4], scheduling with incompatible objects [31, 24], partition into bounded independent sets [29, 11], and m-BoundedColouring [25]. In terms of complexity, m-BoundedColouring is polynomial-time solvable on trees [18], but NP-Hard on co-graphs, interval graphs, and bipartite graphs [11].

As previous suggested, a non-convex relaxation of the problem is provided below:

$$\min t \tag{1}$$
$$\operatorname{rank}(X) = t \tag{2}$$
$$\operatorname{diag}(X) = \bar{e} \tag{3}$$
$$X \succeq 0 \tag{4}$$
$$\sum_{i \in V} X_{ij} \leq m \qquad \forall j \in V \tag{5}$$
$$X_{ij} = 0 \qquad \forall ij \in E \tag{6}$$
$$X_{ij} \in \{0, 1\} \qquad \forall i, j \in V \tag{7}$$

The objective function Equation (1) and first three constraints Equations (2) to (4) come from Proposition 2. Due to Proposition 1, the constraint that every colour class does not exceed m translates to the fourth column (row) cardinality constraint Equation (5). Finally, we enforce mutual-exclusion through constraint Equation (6) and integrality through constraint Equation (7).

In order to develop a convex relaxation, we first relax the 0-1 constraint Equation (7) to a non-negativity constraint. Then we employ similar techniques to rank minimization from [20]. Namely, we think of the trace of a positive semidefinite matrix as the ℓ_1-norm on its eigenvalues and minimize this upper bound. It is known that minimizing the ℓ_1-norm in practice tends to promote sparsity of the eigenvalues and thus tends to yield low rank solutions [21, 32].

However, in the previous program, we require $\operatorname{diag}(X) = \bar{e}$. Thus to allow for minimization, we associate a solution X in the problem Equations (1) to (7) to a pair $(t, Y = tX)$. However, we need to assert that $\operatorname{rank}(Y) \geq 1$, or else the zero matrix is a useless optimal solution. Marecek and Parkes used Proposition 3 without proof[1]:

Proposition 3. *For any integral scheduling matrix X, $nX \succeq J$.*

Proof (Proposition 3). We argue that $h^T(nX - J)h \geq 0$ for all $h \in \mathbb{R}^n$. Since there is some permutation matrix P such that PXP^T is a direct sum of all-ones matrices and J is invariant under $P \cdot P^T$, we may assume without loss of generality that $X = \oplus_{i=1}^{s} J_{d_i}$.

[1] We include our own proof of Proposition 3 here for completeness.

For any $h = \oplus_{i=1}^{s} h^{(i)} \in \mathbb{R}^n$,

$$\begin{aligned} h^T J h &= \left(\sum_{j=1}^{n} h_j \right)^2 \\ &\leq s \sum_{i=1}^{s} \left(\sum_{j=1}^{d_i} h_j^{(i)} \right)^2 && \text{Cauchy-Schwartz Inequality} \\ &\leq n \sum_{i=1}^{s} [h^{(i)}]^T J_{d_i} [h^{(i)}] && s \leq n \\ &= n h^T X h. \end{aligned}$$

By Proposition 3, we see that for sufficiently large t, $Y = tX \succeq J$. Moreover, $Y \succeq J$ implies that $\lambda_i(Y) \geq n$, hence adding this constraint would enforce that $\text{rank}(Y) \geq 1$. This brings us to the SDP relaxation due to Marecek and Parkes for m-BoundedColouring:

$$\min t \tag{8}$$

$$\text{diag}(X) = t\bar{e} \tag{9}$$

$$X \succeq J \tag{10}$$

$$\sum_{i \in V} X_{ij} \leq tm \qquad \forall j \in V \tag{11}$$

$$X_{ij} = 0 \qquad \forall ij \in E \tag{12}$$

$$X_{ij} \geq 0 \qquad \forall i, j \in V \tag{13}$$

2.2 Initial Assignment

In many applications, there is already an existing time table which we must respect and our goal is to augment this time table with new tasks. This corresponds to be following problem:

Problem 2 (m-Bounded Colouring with Pre-Colouring). Given an undirected graph $G = (V, E)$, $m \in \mathbb{Z}_+$, a subset $V' \subseteq V$, and an m-bounded colouring c' of $G[V']$, output a m-bounded colouring $c : V \to \mathbb{Z}_+$ such that every colour class C' of c' satisfies $\forall i, j \in C', c(i) = c(j)$.

This problem is NP-Hard even on trees [10]. However, there are fixed parameter tractable algorithms (FPT) [11, 10]. We can extend the SDP relaxation for m-BoundedColouring for this problem, but a formulation with fewer constraints can be achieved by considering a generalization:

Problem 3 (m-Budgeted Colouring). Given an undirected graph $G = (V, E)$, $m \in \mathbb{Z}_+$, and a weight function $w : V \to \mathbb{R}_+$, return a proper vertex-colouring such that every colour class C satisfies $\sum_{i \in C} w(i) \leq m$.

m-BudgetedColouring is clearly a generalization of m-BoundedColouring as it contains the latter as a special case. We are also able to handle pre-assignments by combining every vertex from a pre-colour class C into a new super vertex which has weight $|C|$. Every vertex without a pre-colouring has weight 1. This leads to the following SDP relaxation:

$$\min t \tag{14}$$
$$\operatorname{diag}(X) = t\bar{e} \tag{15}$$
$$X \succeq J \tag{16}$$
$$\sum_{i \in V} w(i) X_{ij} \leq tm \qquad \forall j \in V \tag{17}$$
$$\sum_{j \in V} w(j) X_{ij} \leq tm \qquad \forall i \in V \tag{18}$$
$$X_{ij} = 0 \qquad \forall ij \in E \tag{19}$$
$$X_{ij} \geq 0 \qquad \forall i, j \in V \tag{20}$$

The only difference of this formulation from that of m-BoundedColouring are the constraints Equation (17), Equation (18). Instead of the simple cardinality constraints, we now have weighted budget constraints. Moreover, since the weight function is no longer symmetric, we need to add constraints to both the rows and columns.

2.3 Simple Timetabling

Imagine a school with m lecture rooms, as well as some lectures to schedule, some of which are mandatory for all students and thus cannot happen simultaneously. Our current model does not take into account the equipment and features of the m rooms, nor the needs of each lecturer. SimpleTimetabling, defined in Appendix A, is a more realistic model. We omit the details for two reasons: all new constraints are linear and the lack of computational data on the effectiveness of these relaxations.

3 Algorithms

3.1 Solving the SDPs

There has been extensive work on (approximately) solving semidefinite programs [35, 13, 50, 38, 45, 23, 49]. We illustrate the ideas behind a first-order method based on the alternating-direction method of multipliers (ADMM) on an augmented Lagrangian.

Consider the primal-dual pair

$$\begin{aligned} \min\langle C, X\rangle \qquad & (P) \\ \mathscr{A}(X) &= b \\ \mathscr{A}_B(X) &\geq d \\ X &\succeq 0 \end{aligned}$$

$$\begin{aligned} \max b^T y + d^T v \qquad & (D) \\ \mathscr{A}^*(y) + \mathscr{A}_B^*(v) + S &= C \\ S &\succeq 0 \\ v &\geq 0 \end{aligned}$$

This is able to accommodate for all of Marecek and Parkes' relaxations. The augmented Lagrangian of the dual is then

$$\begin{aligned} L_\mu(X, y, v, S) := &- b^T y - d^T v \\ &+ \langle X, \mathscr{A}^*(y) + \mathscr{A}_B^*(v) + S - C\rangle \\ &+ \frac{1}{2\mu}\|\mathscr{A}^*(y) + \mathscr{A}_B^*(v) + S - C\|_2^2. \end{aligned}$$

Given an initial feasible solution $\bar{X}$ and steps-size $\bar{\mu}$ (perhaps from a heuristically obtained colouring), we iterative towards an optimal solution within a given precision $\epsilon > 0$. First, compute matching dual variables $\bar{y}, \bar{v}, \bar{S}$. The idea is to then update $\bar{y}, \bar{S}, \bar{v}$ using $L_{\bar{\mu}}$ in alternating fashion:

1) $\bar{y} := \operatorname{argmin}_y L_{\bar{\mu}}(\bar{X}, y, \bar{v}, \bar{S})$.
2) $\bar{v} := \operatorname{argmin}_{v \geq 0} L_{\bar{\mu}}(\bar{X}, \bar{y}, v, \bar{S})$
3) $\bar{S} := \operatorname{argmin}_{S \succeq 0} L_{\bar{\mu}}(\bar{X}, \bar{y}, \bar{v}, S)$

We then choose a new step-length $\bar{\mu} \geq 0$, update $\bar{X}$ according to the new values of $\bar{y}, \bar{v}, \bar{S}, \bar{\mu}$, and repeat. Notice that we reduced a moderately difficult problem to three relatively simpler ones. We refer the interested reader to [49] for convergence properties of such methods as well as implementational details.

Marecek and Parkes also made some observations about the structure of $\mathscr{A}$ for their relaxations of m-BoundedColouring [36]. This enables us to evaluate the augmented Lagrangian and its gradient at a given point in $O(n^2)$ time. This is in contrast to the naive $O(n^6)$ implementation which does not exploit the structure.

3.2 Recovering an Assignment

Since the seminal paper of Karger et al. [26], there has been interest in recovering colourings from semidefinite relaxations. Typically, such algorithms are based on simple randomized rounding of (approximate) optimal solutions. Alternatively, we can consider of solving a sequence of semidefinite programs with decreasing

size. Inspired by iterative rounding in linear programming [30], this method fixes eigenvectors whose eigenvalues are close 0 or 1, and solves a smaller residual program. We defer the interested reader to the paper by Morgenstern et al. [37] for a directly applicable example. There are a myriad of other rounding approaches within literature [6, 40, 5, 1]. Although not directly applicable to scheduling, the ideas upon which they are based are interesting.

Now, approximating the chromatic number of a graph within a factor of $n^{1-\epsilon}$ is NP-hard for all $\epsilon > 0$ [51]. Hence we cannot hope for a rounding algorithm which guarantees solutions close to optimality. However, by leveraging Proposition 4, we can attain an α-*solution* for some $\alpha > 0$. That is, a solution within α of optimality which may violate the linear constraints by an additive error of α. Moreover, the running time is bounded above by a polynomial in n and $\log \frac{1}{\alpha}$.

Proposition 4 ([37]). *Suppose an SDP of the form below has an optimal objective* OPT.

$$\begin{aligned} \min \langle C, X \rangle & \\ \langle A_i, X \rangle \geq b_i & \qquad \forall i \in [m] \\ \mathrm{Tr}(X) \leq d & \\ 0 \preceq X \preceq I & \end{aligned}$$

Given a strictly feasible solution X, there is an efficient algorithm which computes $\tilde{X}$ such that $\operatorname{rank} \tilde{X} \leq d$, $\langle C, \tilde{X} \rangle \leq \mathrm{OPT}$, *and for all $i \in [m]$,*

$$\langle A_i, \tilde{X} \rangle \geq b_i - \max_{S \subseteq [m]} \sum_{j=1}^{\lfloor \sqrt{2|S|}+1 \rfloor} \sigma_j(S),$$

where $\sigma_j(S)$ is the j-th largest singular of the average of matrices $\frac{1}{|S|} \sum_{i \in S} A_i$ for any $S \subseteq [m]$.

While the feasibility violation can be difficult to reason about, each singular value is bounded above by a constant and the summation ranges over at most $\sqrt{2m} + 1$ values.

4 Practical Results

We now summarize the computational tests by Marecek and Parkes on the practicality of their relaxations. Our primary interest is the strength of the lower bounds provided by the SDP relaxations but we also compare the running times of several solvers.

4.1 Random Graphs

Consider random graphs from the class $G\left(n, \frac{1}{2}\right)$, where each unordered pair of vertices are adjacent with independent probability $\frac{1}{2}$. In general, the authors

generated 100 random graphs for each n and plotted the mean lower bound, upper bound, running time, etc.

First, Marecek and Parkes demonstrate the improved strength of the lower bound for m-BoundedColouring as m decreases. The authors used integer programming to compute the size of the largest colour class C in an optimal colouring. Then, they computed SDP lower bounds for the $(C-1), (C-2), (C-3)$-bounded colourings of the same graph as well as upper bounds by rounding the SDP relaxation. In Figure 1, the true value is plotted on a solid line, while a semi-transparent region spans the lower and upper bounds.

Second, let us consider the efficiency of the optimizations Marecek and Parkes developed in Section 3 in solving their relaxations by exploiting the structure of the instances. Figure 2 presents the average running on relaxations for $G\left(n, \frac{1}{2}\right)$ for 100 samples per number n of vertices. Run times are plotted for a standard interior point method (IPM) as well as the optimized augmented Lagrangian method (AugLag).

Next, Marecek and Parkes demonstrated that the run time of the augmented Lagrangian method (AugLag) is roughly linear to the number of vertices when applied to relaxations for $G\left(n, \frac{1}{2}\right)$. Figure 3 depicts this relation even for relatively large instances.

4.2 Timetabling Benchmarks

Marecek and Parkes attempt to demonstrate the effectiveness of their relaxations in practice for conflict graphs from popular timetabling benchmarks. They extracted conflict graphs from both the International Timetabling Competition 2007 [7, 12, 15] as well as the Toronto Examination Timetabling Benchmark [16, 39]. However, the data for the International Timetabling Competition has been labeled without explanation (Table 1). Moreover, there is incomplete data for the Toronto Examination Timetabling Benchmark (Table 2). As presented in [36], it is difficult to discern whether the data actually support the authors' claims.

4.3 Kneser Graphs & Forbidden Intersection Graphs

Marecek and Parkes acknowledged the weakness of their relaxations on certain graphs. A *Kneser graph* $K(n,k)$ for $n > k > 1$ has $\binom{n}{k}$ vertices, corresponding to subsets of $[n]$ with cardinality k. Two subsets are adjacent if and only if they are disjoint. Lovasz showed that the chromatic number of $K(n,k)$ is precisely $n-2k+2$, even though $K(n,k)$ has no triangles for $n > 3k$ [33]. A similar class of graphs are the *forbidden intersection graphs* $F(m,\gamma)$ for $m \geq 1$ and $0 < \gamma < 1$ such that $(1-\gamma)m$ is an even integer. The vertices of $F(m,\gamma)$ are the binary strings of length m and two strings are adjacent if and only if they differ in exactly $(1-\gamma)m$ bits.

It is known that the Lovasz theta number is a poor lower bound on the chromatic number of Kneser graphs [26] as well as forbidden intersection graphs [17]. Suppose C is the size of the largest colour class in an optimal colouring of a

graph. Let χ^{-k} denote the $(C-k)$-bounded chromatic number and X^{-k} the lower bound attained through Marecek and Parkes' relaxation. Table 3 shows that $X^{-k} \in O(1)$ as n grows, while $\chi^{-k} \in \Theta(n)$. Marecek and Parkes observed that this may be due to the large difference between clique and chromatic numbers in both Kneser and forbidden intersection graphs, an issue which is not present for conflict graphs encountered in timetabling applications.

5 Conclusion

In this paper, Marecek and Parkes explored the limits of SDP relaxations in timetabling. The authors cleverly combined properties of integral scheduling matrices to form relaxations that they expect to perform well in practice. However, computational results from popular timetabling benchmarks either contain incomplete entries or have unexplained labels. This is very detrimental towards the strength of their argument.

It is noteworthy that the colour assignment is not directly represented in these relaxations but rather in terms of colour classes. This is sufficient for SimpleTimetabling but not for other objectives in timetabling which require explicit time-slot assignments. The authors posed the problem of modeling more complex problems as future work. Another direction of interest is whether these relaxations are the best possible in polynomial time assuming the unique games conjecture [27]. Yet another area of interest is the use of life-and-project operators in scheduling [5].

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A Additional Problems

Problem 4 (Simple Timetabling). Given

- a graph $G = (V, E)$,
- $m \in \mathbb{Z}_+$,
- vertex weights $w : V \to \mathbb{R}_+$,
- a capacity demand function $p : V \to \mathbb{R}$,
- a capacity supply function $r : [m] \to \mathbb{R}$,
- a number of features $f \in \mathbb{Z}_+$,
- a set of feature demand of events $F \subseteq V \times [f]$,
- a set of feature supply of rooms $G \subseteq [m] \times [f]$,

output a m-budgeted colouring $c : V \to \mathbb{Z}_+$ such that

(i) for every colour class C and distinct capacity μ in r,

$$\sum_{i \in C : p(i) \geq \mu} w(i) \leq |\{m' \in [m]\} : r(m') \geq \mu|$$

(ii) for each colour class C and feature $f' \in [f]$,

$$\sum_{i \in C : (i, f') \in F} w(i) \leq |\{m' \in [m] : (m', f') \in G\}|$$

B Computational Results

Fig. 1. Average lower bounds, upper bounds, and true value of colour classes for bounded colourings.

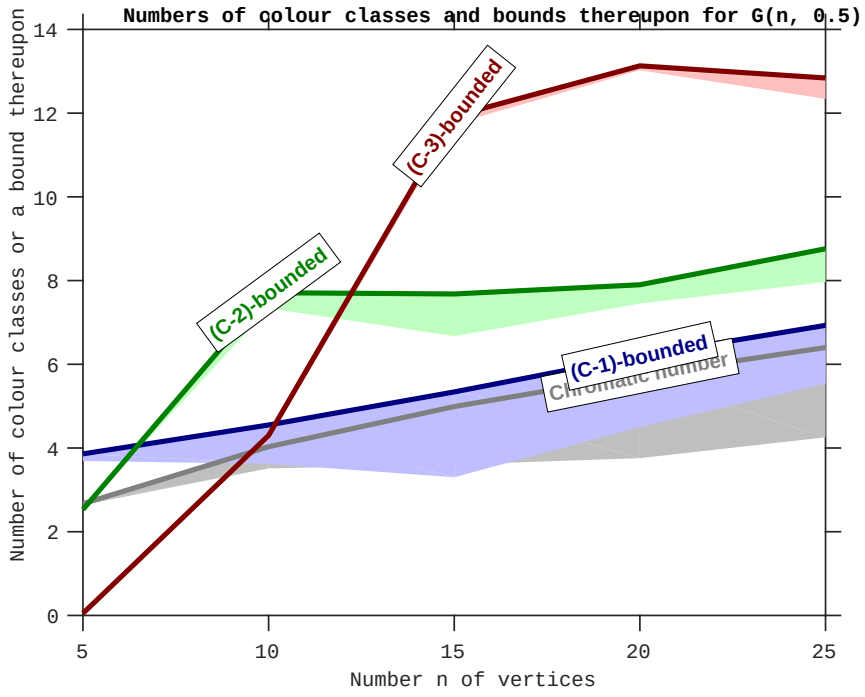

Fig. 2. Average run times for a naive interior point method compared to that of the optimized augmented Lagrangian method.

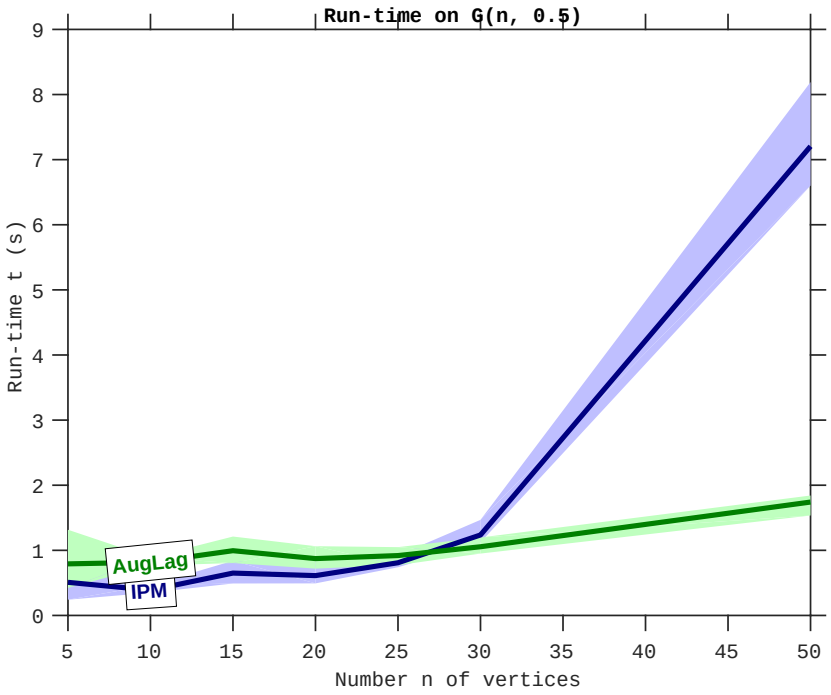

Fig. 3. Average run times when solving relaxations using the optimized augmented Lagrangian method.

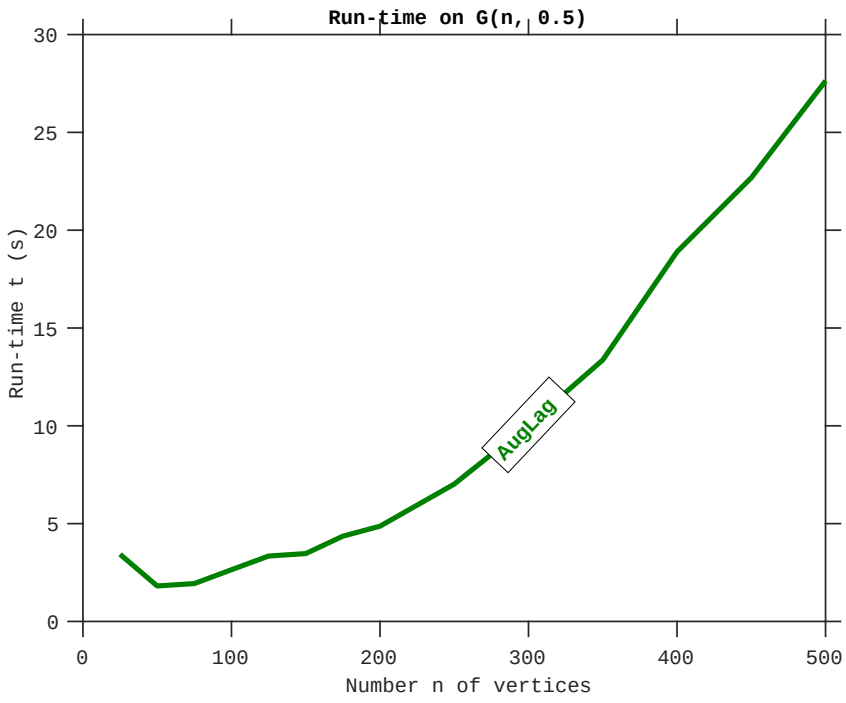

Table 1. International Timetabling Competition data

Graph	X	Runtime	X^m	Runtime	Rounded
comp01.course	4.00	4 s	5.00	0 s	7
comp02.course	5.98	2 s	6.00	7 s	12
comp03.course	6.94	1 s	7.00	7 s	14
comp04.course	4.98	1 s	5.00	3 s	12
comp05.course	8.00	1 s	7.99	3 s	14
comp06.course	5.99	3 s	6.00	8 s	14
comp07.course	6.00	5 s	6.55	21 s	17
comp08.course	6.99	2 s	6.98	7 s	11
comp09.course	4.99	1 s	5.00	5 s	10
comp10.course	6.00	3 s	6.39	11 s	16
comp11.course	5.00	1 s	6.00	0 s	8
comp12.course	9.91	4 s	9.96	15 s	18
comp13.course	5.98	1 s	6.00	7 s	8
comp14.course	6.00	2 s	6.00	10 s	14
comp15.course	6.94	1 s	7.00	7 s	15
comp16.course	6.00	4 s	5.99	9 s	15
comp17.course	5.98	3 s	6.00	14 s	12
comp18.course	4.99	1 s	5.22	1 s	8
comp19.course	6.00	1 s	6.00	5 s	11
comp20.course	6.00	4 s	6.37	14 s	13
comp21.course	8.00	2 s	8.00	14 s	13

Table 2. Toronto Examination Timetabling Competition and International Timetabling Competition data.

Graph Parameter	X^0	χ^0	X^{-1}	χ^{-1}	X^{-2}	χ^{-2}	X^{-3}	χ^{-3}
comp01.event	4.00	4	4.00	4		5		5
comp02.event		6		6		6		6
comp03.event	7.00	7	7.00	7	7.00	7	7.00	7
comp04.event		5		5	5.00	5		7
comp05.event		8		8	9.00	9	10.80	11
comp06.event		6		6		6		6
comp07.event		6		7		7		7
comp08.event		7		7		7		7
comp09.event		5		6	5.43	7	5.85	6
comp10.event		6		6		6		7
comp11.event	5.00	5	5.00	5		5		5
comp12.event	10.00	10	10.00	10		10		10
comp13.event	6.00	6	6.00	6	6.00	6		6
comp14.event		6		6		7		6

Table 3. Computational results in Kneser and forbidden intersection graphs.

Graph	X^0	χ^0	X^{-1}	χ^{-1}	X^{-2}	χ^{-2}	X^{-3}	χ^{-3}
$K(5,2)$	2.50	3	3.33	4	5.00	5	10.00	1
$K(6,2)$	3.00	4	3.75	4	5.00	5	7.50	8
$K(7,2)$	3.50	5	4.20	5	5.25	6	7.00	7
$K(8,2)$	4.67	6	5.60	6	7.00	7	9.33	10
$F(6,0.50)$	2.00	2	2.03	3	2.06	3	2.13	3
$F(6,0.67)$	6.40	7	7.11	8	8.00	8	9.14	10
$F(6,0.83)$	2.00	2	2.03	3	2.06	3	2.13	3
$F(6,1.00)$	2.00	2	2.00	2	2.06	3	2.13	3